

Which dark energy can structure growth see? A response-kernel classification of geometric dark energy

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Abstract

A dark-energy model whose density is fixed by a geometric construction—a horizon length, a curvature scalar, a nonlocal functional of the metric—has no independent perturbation sector: its density contrast is *inherited* from the gravitational potential through whatever functional defines it. We show that the observability of such a model in the linear structure-growth rate $f\sigma_8$ is controlled by a single quantity: the ultraviolet scaling of the kernel through which the dark-energy density responds to the potential. Because the density perturbation is always a linear functional of the metric perturbation, geometric dark-energy models partition exhaustively by this kernel’s large- k behaviour into a small number of classes. Local constructions (Hubble, Ricci, Granda–Oliveros, and—we emphasize—the apparent horizon) have a delta-function kernel, cluster like an ordinary $c_s^2 = 1$ fluid, and remain growth-informative. Light-cone constructions (the future event horizon and the particle horizon) have a one-dimensional oscillatory kernel; phase-averaging of the potential along the ray suppresses their density contrast as $\delta_{\text{DE}}/\delta_{\text{m}} \sim 1/k^3$, so they carry no growth information beyond the background expansion. Smooth-nonlocal constructions are smoother still. We give the criterion, prove the partition is exhaustive, identify the future event horizon as the unique cutoff whose growth channel is *unconditionally* closed, and delineate the two boundary subtleties—acoustic resonance on the particle-horizon ray, and a crossover scale for cutoffs with a small local admixture—that separate the exact statement from its observational version. The result reorganizes a large and fragmented model space around a single computable diagnostic and tells an observer, in advance, which geometric dark-energy models a growth measurement can and cannot constrain.

1 Introduction

Dark energy is increasingly probed not only through the background expansion $H(z)$ but through the rate at which cosmic structure grows, $f\sigma_8(z)$. The premise of that program is that growth carries information the expansion does not: two models with identical $H(z)$ can differ in how dark energy clusters, and $f\sigma_8$ sees the difference. For a broad and popular class of models, however, that premise fails—and it fails for a structural reason that can be stated once and checked model by model.

The class in question is what we will call *geometric dark energy*: models in which the dark-energy density is not an independent dynamical field but is fixed by a construction built from the spacetime geometry. Holographic dark energy, $\rho_{\text{DE}} \propto L^{-2}$ with L an infrared cutoff length, is the best-known example, and its generalized-entropy variants (Barrow, Tsallis, Rényi, Kaniadakis) form a large post-DESI literature. But the category is wider: it includes Ricci and Granda–Oliveros dark energy, apparent-horizon and thermodynamic constructions, and nonlocal-gravity models in which the dark-energy density is a nonlocal functional of the curvature.

For every such model the same fact holds: the dark-energy density perturbation is not free. If $\rho_{\text{DE}} = f(L[g])$ for some geometric length functional L of the metric g ,

then

$$\delta_{\text{DE}} = \frac{d \ln f}{d \ln L} \frac{\delta L}{L}, \quad (1)$$

and the entire scale dependence of δ_{DE} lives in δL , the perturbation of the geometric construction. The prefactor—where all the entropy physics sits—is a bounded, scale-independent number. Whether growth can see the model is therefore not a question about entropy or equation of state; it is a question about δL , and specifically about *how δL responds to the gravitational potential as a function of scale*.

This paper makes that question precise and answers it in general. We show (Sec. 2) that δL is always a linear functional of the metric perturbation, so that in Fourier space $\delta L_k = \int \mathcal{K}(k; \cdot) \Phi$, and that the observability of the model is fixed by the large- k scaling of the kernel $\mathcal{K}$. We prove (Sec. 3) that geometric dark-energy models partition exhaustively into a small number of classes by this scaling, and that only the local class remains growth-informative at observable scales. We identify (Sec. 4) the future event horizon as the unique construction whose channel is unconditionally closed, and we treat (Sec. 5) the two boundary cases—particle-horizon acoustic resonance and mixed cutoffs—that mark the edge of the clean statement. The holographic closed-channel result established for event-horizon models elsewhere [1] is the central, cleanest member of the classification given here.

2 The response kernel

We work in the conformal-Newtonian gauge with metric potential Φ , on sub-horizon scales $k \gg aH$ where the observations live, and to linear order. Matter perturbations obey the standard growth equation, whose only cosmological input is $H(a)$; two models with the same expansion history give the same matter growth unless dark energy itself clusters and back-reacts through the Poisson equation,

$$-\frac{k^2}{a^2}\Phi = 4\pi G [\rho_m \delta_m + \rho_{\text{DE}} \delta_{\text{DE}} (1 + 3c_{\text{eff}}^2)]. \quad (2)$$

The dark-energy term is the whole question. Its size relative to the matter term is governed by δ_{DE} , which by Eq. (1) is set by δL .

δL is a linear functional of Φ . A geometric length L is a functional of the metric. Its first-order perturbation is therefore a linear functional of the metric perturbation: schematically

$$\delta L(x, t) = \int \mathcal{K}(x, t; x', t') \Phi(x', t') d^4 x', \quad (3)$$

for a kernel $\mathcal{K}$ determined entirely by how L is built from g . In Fourier space $\delta L_k = \tilde{\mathcal{K}}(k) \Phi_k$ (times a possible integral along a curve or over a region, made explicit below). Sub-horizon, the potential is sourced by matter, $\Phi_k \sim -4\pi G a^2 \rho_m \delta_m / k^2$, so the density contrast carries the Poisson factor $1/k^2$ together with the ultraviolet scaling of the kernel,

$$\frac{\delta_{\text{DE}}}{\delta_m} \sim \frac{1}{k^2} S(k), \quad S(k) \equiv \frac{\tilde{\mathcal{K}}(k)}{\tilde{\mathcal{K}}_{\text{local}}}, \quad (4)$$

with S normalized to unity for a purely local response. The organizing claim of this paper is that the growth-observability of a geometric dark-energy model is fixed by $S(k)$ —what the geometry adds beyond the universal pressure (Jeans) floor $1/k^2$. Section 3 makes both factors precise: the floor is the same for every construction (Lemma 1), while $S(k)$ takes one of a few forms set by the type of geometric functional (Lemma 2). We caution that the local class does *not* follow from a naive derivative count on $\tilde{\mathcal{K}}$; its value is set by the pressure floor, as the proof makes explicit.

$S(k)$ is gauge-invariant. Although δL for a nonlocal length is gauge-dependent—it integrates the metric perturbation over a curve or region, and a coordinate change shifts the integrand—the ultraviolet scaling of $S(k)$ is not. Two facts secure this. First, the potential appearing in Eq. (3) is the Newtonian-gauge Φ , which equals the Bardeen invariant Φ_A ; the geometric object over which it is integrated (a null ray, a causal region, the support of $\square^{-1}$) is coordinate-independent, so the leading large- k behaviour of $\tilde{\mathcal{K}}$ is a property of the invariant pair

$(\Phi_A, \text{geometry})$, not of the slicing. Second, S is defined as a *ratio* to the local response, and any residual gauge contribution to δL enters at $\mathcal{O}(aH/k)$ relative to the leading term—the same order in every gauge—so it cancels in the ratio to the order at which $S(k)$ is evaluated. A gauge transformation can therefore rescale δL but cannot change the exponent of $S(k)$; the local/nonlocal boundary of Lemma 2, which depends only on that exponent, is a gauge-invariant statement. We verified this explicitly for the light-cone class by recomputing the leading scaling with Φ_A in place of the Newtonian Φ : the $1/k$ is unchanged.

3 The classification

The kernel $\tilde{\mathcal{K}}(k)$ is fixed by the *type* of geometric object L is. There are only a few types, distinguished by the support of $\mathcal{K}$ in Eq. (3), and they exhaust the possibilities because every geometric L is one of them.

Definition 1 (Kernel classes). (*Table 1 places the standard constructions in these classes.*) A geometric length functional L is

- (**L**) local if $\mathcal{K}$ is a delta function (with possible derivatives): $\delta L = c(t)\Phi + \dots$. Then $\tilde{\mathcal{K}} \sim k^0$.
- (**N**) null-line if $\mathcal{K}$ has one-dimensional support along a null ray: $\delta L_k = \int_{\text{ray}} \Phi_k(u) e^{iku} du$. Then $\tilde{\mathcal{K}} \sim 1/k$ for a smooth envelope.
- (**S**) smooth-nonlocal if $\mathcal{K}$ is a rapidly decaying spatial window (e.g. a Gaussian average over scale R , $\tilde{\mathcal{K}} \sim e^{-k^2 R^2}$).
- (**I**) inverse-box nonlocal if L involves $\square^{-1}$ acting on curvature, so $\tilde{\mathcal{K}} \sim 1/k^2$.

Theorem 1 (Exhaustive growth partition). *Let dark energy be geometric, $\rho_{\text{DE}} = f(L[g])$ with $d \ln f / d \ln L$ bounded and nonzero. Then its sub-horizon density contrast relative to matter scales as*

$$\frac{\delta_{\text{DE}}}{\delta_m} \sim \begin{cases} 1/k^2 & \text{class (L): clusters, growth-informative} \\ 1/k^3 & \text{class (N): growth-empty} \\ \ll 1/k^n \forall n & \text{class (S): super-smooth} \\ 1/k^4 & \text{class (I): growth-empty} \end{cases} \quad (5)$$

and every geometric L is growth-informative or growth-empty according to the dichotomy of Lemma 2; the four rates are those of the constructions in current use.

The classification rests on a decomposition of the density response into a universal part and a geometry-dependent part.

Lemma 1 (Jeans floor and suppression factor). *For geometric dark energy with $\rho_{\text{DE}} = f(L[g])$, the sub-horizon density contrast factorizes as*

$$\frac{\delta_{\text{DE}}}{\delta_{\text{m}}} \sim \underbrace{\frac{1}{k^2}}_{\text{Jeans floor}} \times S(k), \quad (6)$$

where $S(k)$ is the ultraviolet scaling of the kernel $\tilde{\mathcal{K}}(k)$ through which δL responds to the potential, normalized so that a purely local response gives $S = 1$.

Proof. Two facts combine. (i) *The floor.* We take as the defining property of *geometric* dark energy that its density is a functional of the metric with no independent matter or scalar degree of freedom—no separate kinetic term in the action carrying its own momentum. For such a component the rest-frame sound speed is not a free Lagrangian parameter; the prescription that renders $\rho_{\text{DE}} = f(L[g])$ well-defined ties its pressure perturbation to its density perturbation through the same geometric relation, fixing the effective sound speed to the smooth branch $c_{\text{eff}}^2 = 1$. A component with $c_{\text{eff}}^2 = 1$ has its sound horizon equal to the Hubble radius, so sub-horizon its perturbations are pressure-supported and Jeans-suppressed, $\delta_{\text{DE}}/\delta_{\text{m}} \sim (aH/k)^2 = 1/k^2$ (in units $aH = 1$) [2, 3]. This is the universal floor for the geometric class, independent of which construction is used. We flag the one way out explicitly: a functional that implicitly reintroduces an independent momentum flux—for instance by making L depend on the matter velocity field, not the metric alone—would carry its own $c_{\text{eff}}^2 \neq 1$ and fall outside this class; our floor, and hence the partition, is asserted for metric functionals, which is the case of interest and covers every construction in Table 1. (ii) *The factor.* By Eq. (1), $\delta_{\text{DE}} = (d \ln f / d \ln L) \delta L / L$ with the prefactor bounded, and by Eq. (3) $\delta L_k = \tilde{\mathcal{K}}(k) \Phi_k$. Any suppression of $\tilde{\mathcal{K}}$ beyond the local (delta-function) response multiplies the floor: writing $S(k) \equiv \tilde{\mathcal{K}}(k) / \tilde{\mathcal{K}}_{\text{local}}$, the response is the floor times $S(k)$. A local response has $S = 1$ and saturates the floor; a nonlocal one has $S(k) \rightarrow 0$ and lies below it. The two statements are independent—(i) is a fluid pressure result, (ii) a functional-analytic scaling—so their product is Eq. (6). $\square$

Lemma 2 (Locality dichotomy). *On the homogeneous background the response $\Phi \mapsto \delta L$ is linear and translation-invariant, hence diagonal in Fourier space, $\delta L_k = \tilde{\mathcal{K}}(k) \Phi_k$, and the suppression factor $S(k) = \tilde{\mathcal{K}}(k) / \tilde{\mathcal{K}}_{\text{local}}$ has a well-defined ultraviolet order. Then:*

- (i) *If $\mathcal{K}$ contains a local (delta-supported) component— L depending on the metric and finitely many derivatives at the point—that component dominates the ultraviolet, $S(k) \rightarrow \text{const}$ or grows, so $\delta_{\text{DE}}/\delta_{\text{m}} \gtrsim 1/k^2$: the dark energy clusters at or above the Jeans floor and remains growth-informative.*

- (ii) *If $\mathcal{K}$ is genuinely nonlocal—an integrable kernel with no delta component—then by the Riemann–Lebesgue lemma $\tilde{\mathcal{K}}(k) \rightarrow 0$, so $S(k) \rightarrow 0$ and $\delta_{\text{DE}}/\delta_{\text{m}} = o(1/k^2)$: the dark energy is smoother than the Jeans floor and is growth-empty.*

The boundary $S \sim k^0$ is therefore exactly the local/nonlocal divide.

Proof. Linearity and translation invariance on an FRW background give the Fourier diagonalization directly. For (i), a delta-supported kernel has $\tilde{\mathcal{K}}(k) = \sum_n c_n (ik)^n$, which does not vanish at large k ; the leading term sets S , and the density response is bounded below by the $c_{\text{eff}}^2 = 1$ Jeans floor of Lemma 1(i), since a local $\delta_{\text{DE}} \sim \Phi$ is a genuine pressure-supported fluid perturbation (the extra derivative terms of a curvature cutoff enter only at subleading order $(aH/k)^2$ and do not lift the response above the floor’s scaling). For (ii), an L^1 kernel with no atomic part has $\tilde{\mathcal{K}}(k) \rightarrow 0$ by Riemann–Lebesgue, hence $S(k) \rightarrow 0$; the model is strictly smoother than a local one at every large k . $\square$

The dichotomy is the rigorous content: *growth can see a geometric dark-energy model if and only if its density responds to the potential through a locally-supported kernel.* Intuitively: a local response reads the potential at a point, so a small-scale fluctuation registers at full strength; a nonlocal response averages the potential over an extended ray or region, and a small-scale fluctuation—many crests and troughs within the averaging domain—largely cancels itself out. The more spread-out the kernel, the more complete the cancellation, which is why the smooth spatial window (class S) is smoother than the light-cone integral (class N), which is smoother than the local cutoff (class L). Riemann–Lebesgue is the precise form of “averaging an oscillation sends it to zero.” The specific exponents of Theorem 1 refine this by computing the *rate* at which $S(k)$ vanishes for the nonlocal constructions actually in use—a rate that Riemann–Lebesgue alone does not fix. For a light-cone integral, integration by parts on the oscillatory kernel gives $S \sim 1/k$ (a smooth envelope; Sec. 5), hence $1/k^3$. For an inverse d’Alembertian, $S \sim 1/k^2$, hence $1/k^4$. For a smooth spatial window, S decays faster than any power (Paley–Wiener). We do not claim these exhaust all conceivable covariant functionals; they are the constructions the literature employs, and each is placed by direct computation of its kernel order in Table 1.

Proof of Theorem 1. Lemma 1 gives $\delta_{\text{DE}}/\delta_{\text{m}} \sim S(k)/k^2$, and Lemma 2 establishes the dichotomy: $S \rightarrow \text{const}$ for a locally-supported kernel (clustering, growth-informative) and $S \rightarrow 0$ for a nonlocal one (growth-empty). The four entries of Eq. (5) are the values of $S(k)/k^2$ for the constructions in Table 1, using the kernel rates computed after Lemma 2: $S = 1$ (local) gives $1/k^2$; $S = 1/k$ (light-cone) gives $1/k^3$; $S = 1/k^2$ (inverse-box) gives $1/k^4$; a

Table 1: Geometric dark-energy constructions by kernel class. Only class (L) is informative to linear growth at observable scales.

Construction	$L \sim$	Class
Hubble cutoff	$1/H$	(L)
Ricci cutoff	$(\dot{H} + 2H^2)^{-1/2}$	(L)
Granda–Oliveros	$(\alpha H^2 + \beta \dot{H})^{-1/2}$	(L)
Apparent horizon (flat)	$1/H$	(L)
Future event horizon	$a \int_t^\infty dt'/a$	(N)
Particle horizon	$a \int_0^t dt'/a$	(N)
Gaussian-averaged scale	smoothing window	(S)
Nonlocal $\square^{-1}R$	inverse d’Alembertian	(I)

smooth window gives faster than any power. The partition into growth-informative and growth-empty is exhaustive by the dichotomy; the specific exponents catalog the constructions in use rather than claiming to exhaust all covariant functionals. $\square$

Table 1 places the standard constructions. The entry that most often causes confusion is the apparent horizon: in a spatially flat Friedmann–Robertson–Walker background the apparent horizon coincides with the Hubble radius, $L = 1/H$, which is a *local* function of the expansion rate. Apparent-horizon dark energy is therefore in class (L)—it clusters—despite the “horizon” name that might suggest a light-cone construction. Only the future event horizon and the particle horizon, defined by null-geodesic integrals, are class (N).

4 The event horizon is unconditionally closed

Among the growth-empty class (N), the future event horizon is the cleanest case. Its defining ray runs from the observation time into the infinite future, a region that (for accelerating cosmologies) is smoothly dark-energy dominated: the potential envelope along the ray is monotone and non-oscillatory, so the phase-averaging argument applies with no envelope condition at all, and $\delta_{\text{DE}}/\delta_{\text{m}} \sim 1/k^3$ holds for every event-horizon construction—every generalized entropy, every deformation parameter—because the entropy enters only the bounded prefactor in Eq. (1). The particle horizon reaches the same $1/k^3$ but by a slightly stronger route: its ray does cross the oscillatory radiation era, yet the phase-average survives because the ray is null and cannot resonate with the sub-luminal acoustic oscillation (Sec. 5). Class (N) is thus closed throughout, with the event horizon closed *unconditionally* and the particle horizon closed by the light-versus-sound separation. This is the sense in which the holographic closed-channel result [1] is not one result among many but the representative of a class.

It is worth noting that this same cutoff is the one the data independently prefer. Combined expansion-plus-growth analyses disfavor the Ricci and Granda–Oliveros (class (L)) constructions because they overproduce early dark energy, while the future-event-horizon construction remains viable [4]. The cutoff for which the growth channel is provably empty is thus also the cutoff that survives the data—so the closure is not a statement about a disfavored corner of model space but about the observationally central one.

5 Two boundary subtleties

The clean partition of Theorem 1 has two edges where the exact mathematical statement and its observational version diverge. Both are benign, and naming them precisely is part of the result.

Acoustic resonance on the particle-horizon ray.

The particle horizon is class (N), but its ray threads the radiation era, where the potential is not a smooth envelope but oscillates with the acoustic phase. Using the exact radiation-era transfer function $\Phi \propto [\sin x - x \cos x]/x^3$ with $x = c_s k \tau$, the large- x envelope oscillates at frequency $c_s k$, while the null kernel oscillates at γk , where γ is the ray’s propagation speed in the relevant coordinate. Their product beats to zero frequency—defeating the phase-average and degrading the suppression from $1/k^3$ to $1/k^2$ —only if

$$\gamma = c_s, \quad (7)$$

that is, only if the null ray propagates at the acoustic sound speed. It does not: the ray is null, γ is the speed of light, and light is strictly faster than sound in the photon-baryon plasma, $c_s = 1/\sqrt{3} < 1$. The resonance is therefore not a fine-tuned near-miss but a structural impossibility, and a high-precision evaluation of the ray integral with the full transfer function confirms $\tilde{\mathcal{K}} \sim 1/k$ (hence $\delta_{\text{DE}}/\delta_{\text{m}} \sim 1/k^3$), robustly and independently of the ray parametrization. The particle-horizon channel is thus closed for the same structural reason as the event-horizon channel: the only configuration that would re-open it requires sound to overtake light.

Envelope regularity along the ray. The class-(N) scaling $\tilde{\mathcal{K}} \sim 1/k$ was stated for a smooth envelope, which invites the objection that a sharp background feature—a rapid phase transition, a sudden shift in the dominant component—makes $\Phi(u)$ along the ray non-smooth and could, through the boundary terms of the integration by parts, degrade the suppression back toward class (L). It cannot. Integration by parts on $\int \Phi(u) e^{iku} du$ converts each order of smoothness into one power of $1/k$: a jump discontinuity in the envelope contributes a surface term of order $1/k$ (oscillating in k through a phase e^{iku_0}), a kink (derivative discontinuity) contributes $1/k^2$, and a

smooth envelope gives the interior $1/k$. In every case $\tilde{\mathcal{K}} = \mathcal{O}(1/k)$ or smaller. A direct evaluation with an imposed jump and an imposed kink confirms $\tilde{\mathcal{K}} \sim k^{-1}$ unchanged. Lifting the model to class (L) ($S \sim k^0$) would require an $\mathcal{O}(1)$, non-decaying $\tilde{\mathcal{K}}$ —that is, a delta function in Φ itself, a singular potential, which is unphysical for any finite-energy perturbation. The honest statement is therefore that the smooth-envelope assumption can be relaxed to Φ of bounded variation along the ray, under which $\tilde{\mathcal{K}} = \mathcal{O}(1/k)$ and the class-(N) suppression is at worst $1/k^3$; sharp features can only make it faster, never slower, and never local.

Mixed cutoffs and the crossover scale. Class membership is a statement about the *pure* construction. A cutoff with a local admixture, $L = \alpha L_{\text{EH}} + \beta/H$, has $\delta L_k = \alpha \mathcal{O}(1/k) + \beta \mathcal{O}(1)$: the local piece carries no k -suppression and dominates once $k > k_{\text{cross}} \equiv \alpha/\beta$. Any nonzero β therefore returns the model to class (L) asymptotically—but only beyond k_{cross} . For a small admixture ($\beta \ll \alpha$) the crossover lies far above the observable range $k/aH \sim 1\text{--}10^3$, and the model is *effectively* growth-empty across the window a survey can measure. The exact boundary ($\beta = 0$, unconditional $1/k^3$) and the observational boundary (“dominantly null,” k_{cross} beyond reach) thus differ, in exactly the way an exact degeneracy differs from a precision-limited one. We state both.

The comparable-weight case $\alpha \sim \beta$ does not weaken the result, for two reasons. First, it is not a marginal case for the theorem: with $\beta \sim \alpha$ one has $k_{\text{cross}} \sim 1$, so the model is class (L) across essentially the entire observable range—it is *growth-informative*, and the theorem classifies it correctly and unambiguously as such. There is no regime the partition fails to cover; the crossover simply tells us *which* class a mixed cutoff occupies as a function of scale. Second, a comparable local admixture is disfavored on independent grounds: the β/H Hubble-cutoff piece reintroduces exactly the early-dark-energy overproduction that combined expansion-plus-growth data disfavor for the pure Hubble and Ricci cutoffs (Ref. [4]), scaled by β . A model with $\beta \sim \alpha$ inherits an $\mathcal{O}(1)$ fraction of that disfavored behaviour and is constrained by the same data that already disfavor class (L) horizon cutoffs. So the mixed cutoff is either dominantly null (and effectively growth-empty, our case) or carries a substantial local piece (and is both growth-informative *and* background-disfavored)—never a loophole that is simultaneously growth-hidden and observationally viable.

6 Discussion

The classification converts a fragmented question—*does this particular horizon dark energy cluster?*—into a single computable diagnostic: identify the kernel through which the density responds to the potential, read off its ultraviolet scaling, and place the model in Eq. (5). The answer

requires no perturbation code and no fit; it follows from the form of the geometric construction. For an observer, the practical content is a prediction made in advance: a growth survey can constrain class (L) geometric dark energy and cannot constrain classes (N), (S), or (I) beyond what the background already fixes. For the holographic program specifically, it means the event-horizon models that dominate the literature are, as a class, invisible to $f\sigma_8$ —and that the search for a growth signature should be redirected to the local-cutoff constructions, where a signal can in principle exist.

Two limitations bound the scope. First, the framework is linear and sub-horizon; the near-horizon and mildly nonlinear regimes, where the clean k -scalings blur, are left for future work, though the observational weight of $f\sigma_8$ sits at sub-horizon scales where the classification is sharp. We note that nonlinear mode-coupling transfers power between scales and could in principle feed the suppressed dark-energy contrast; but the coupling is through the same potential Φ , whose response is already classified, and the leaked amplitude is bounded by the linear suppression times the (small) nonlinear transfer efficiency—so a class-(N) model cannot acquire an $\mathcal{O}(1)$ clustering signal from nonlinearity, only a higher-order correction to an already-tiny one. A quantitative nonlinear treatment is nonetheless the natural next check. Second, the classification is of *geometric* dark energy—models whose density is inherited from the metric. It says nothing about scalar-field dark energy (quintessence, k-essence), which carries a genuine Lagrangian sound speed and an independent perturbation sector; such models are correctly outside the partition, and the boundary between “geometric” and “field-theoretic” dark energy is itself the natural next question.

Two further caveats deserve explicit mention. The class-(I) inverse-d’Alembertian entry is placed by the leading Fourier scaling $\square^{-1} \sim 1/k^2$ of the nonlocal operator on the background; genuine nonlocal-gravity theories carry their own causal boundary conditions and a richer operator structure, and we classify only the leading UV scaling of the density response, not the full dynamics of such theories—the $1/k^4$ entry should be read as “at least as smooth as class (N),” with the precise treatment of a specific nonlocal-gravity action left to a dedicated analysis. And the $c_{\text{eff}}^2 = 1$ Jeans floor assumes adiabatic initial conditions; a large dark-energy isocurvature mode would alter the initial density scaling before the geometric kernel acts. For geometric dark energy this is not expected—the density is slaved to the metric and inherits the adiabatic mode—but a model with a genuinely independent isocurvature initial condition would sit outside the floor’s assumptions, worth noting for completeness.

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